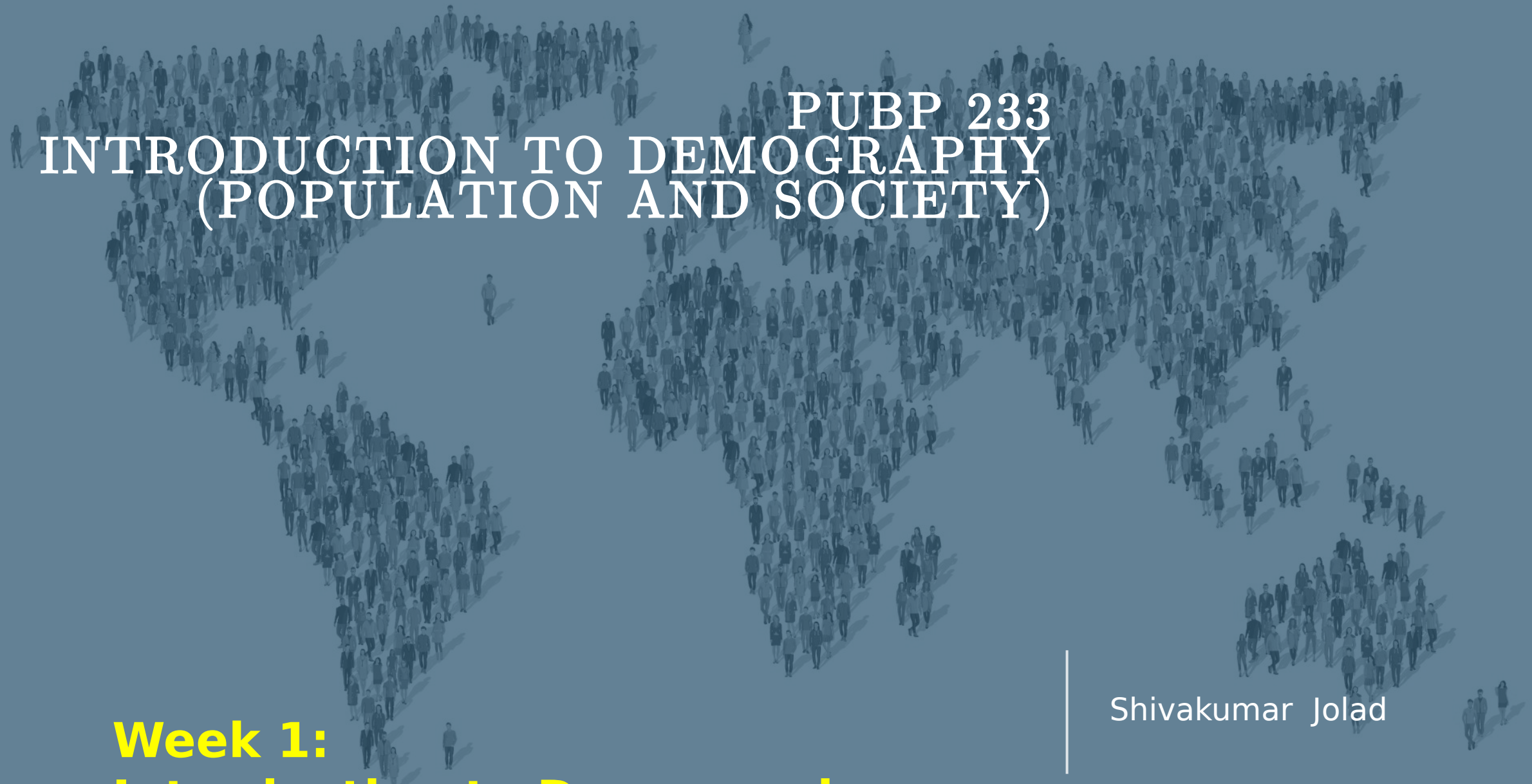
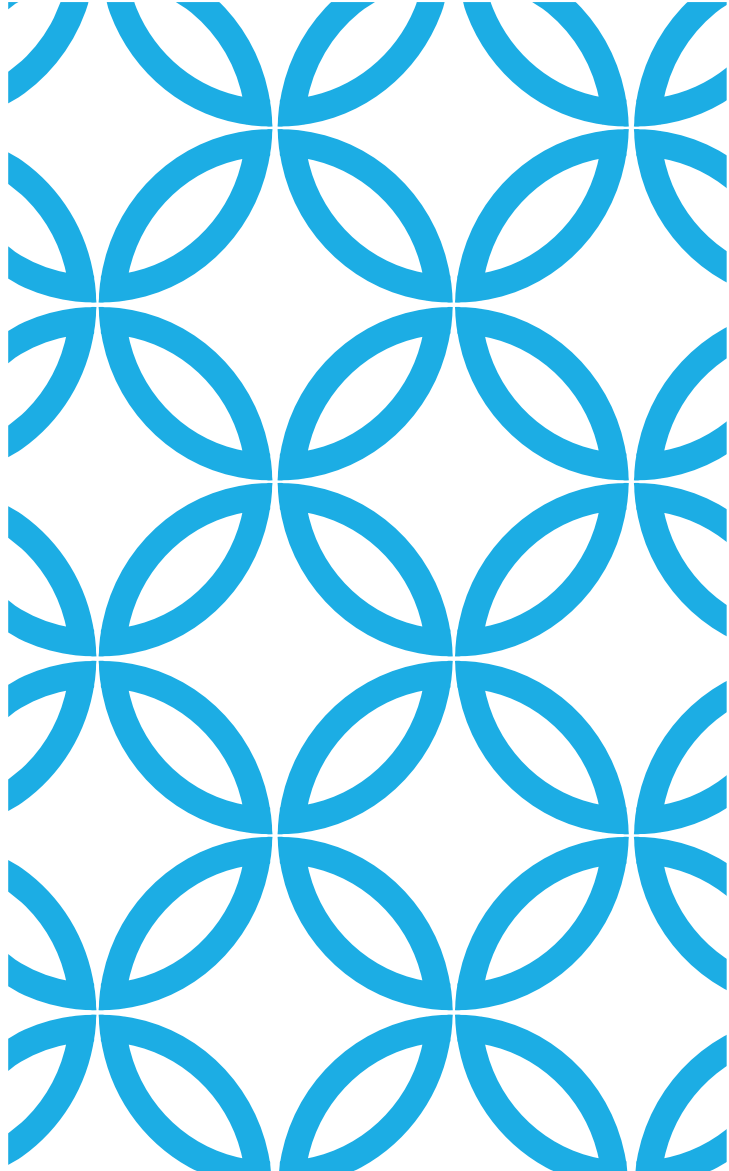




PUBP 233 INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

Week 1:
Introduction to Demography,
Course Information

Shivakumar Jolad



COURSE INFO AND POLICIES

BASIC INFO

COURSE CODE: PUBP233

CREDITS: 3

PROGRAM: Undergraduate

COURSE STATUS (CORE/ ELECTIVE): Core

READ THE PICTURE



Pic: Habib Hasan, Public Health Foundation of India



POPULATION

Government

Economy

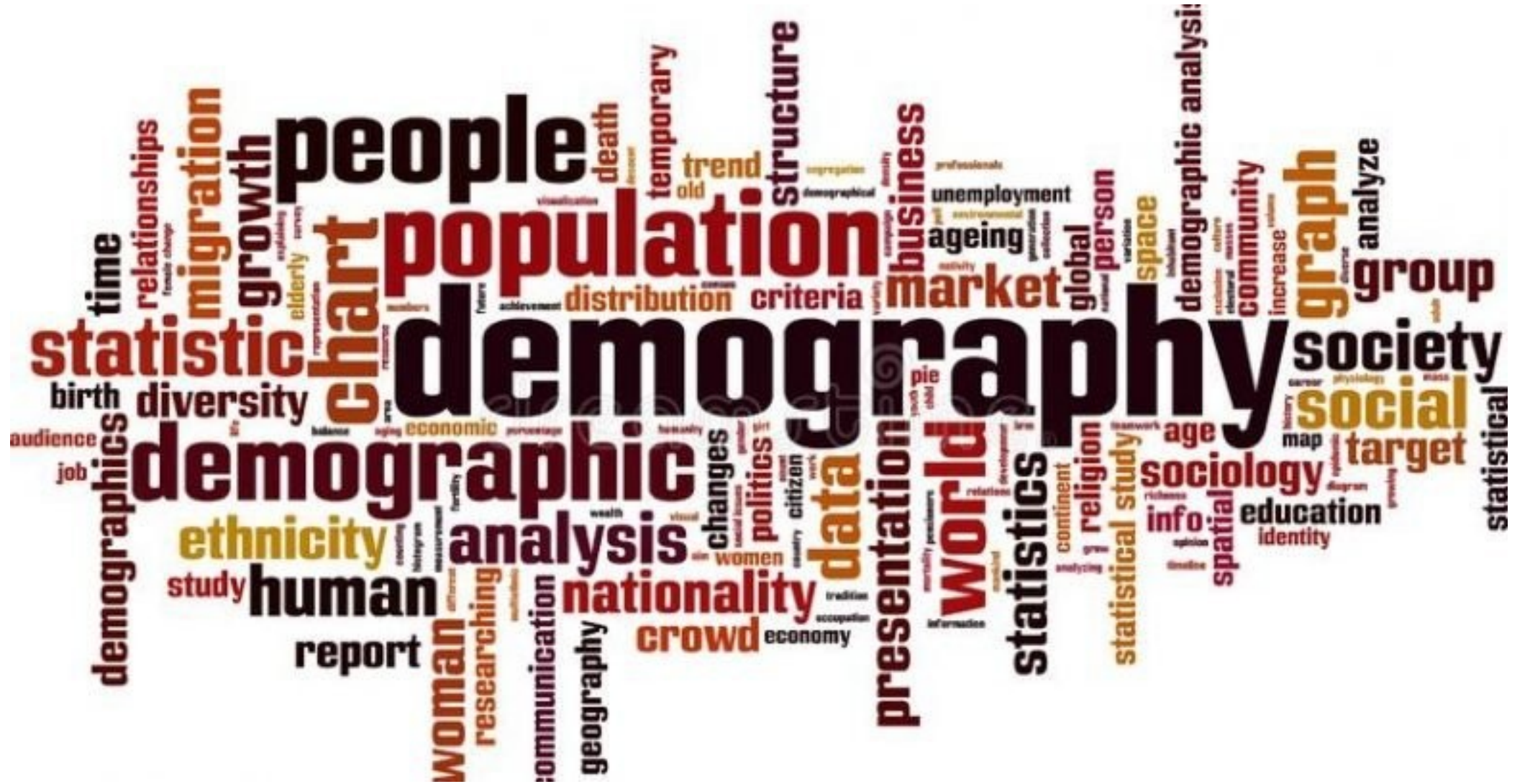
Markets

Society

Education

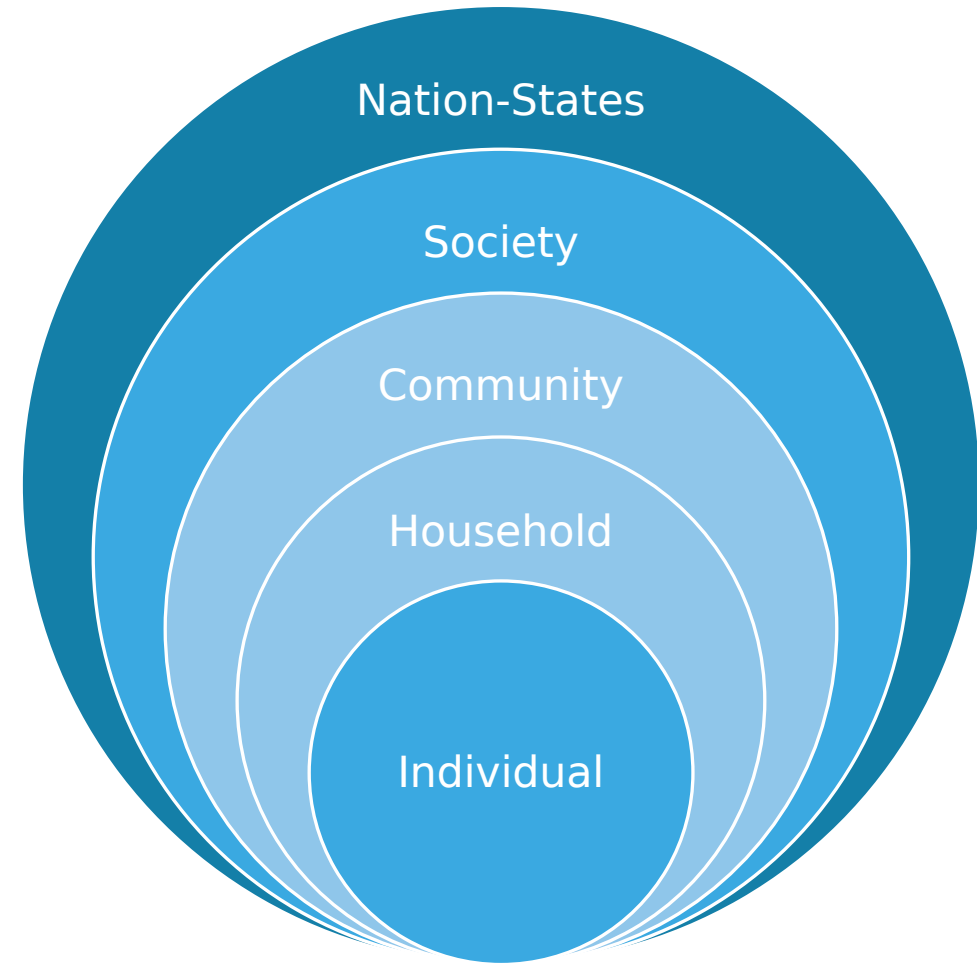
Health

Religion



<https://themarketsresearch.com/what-are-demographics/>

POPULATION DIVISIONS



POPULATION GROUPINGS

Age

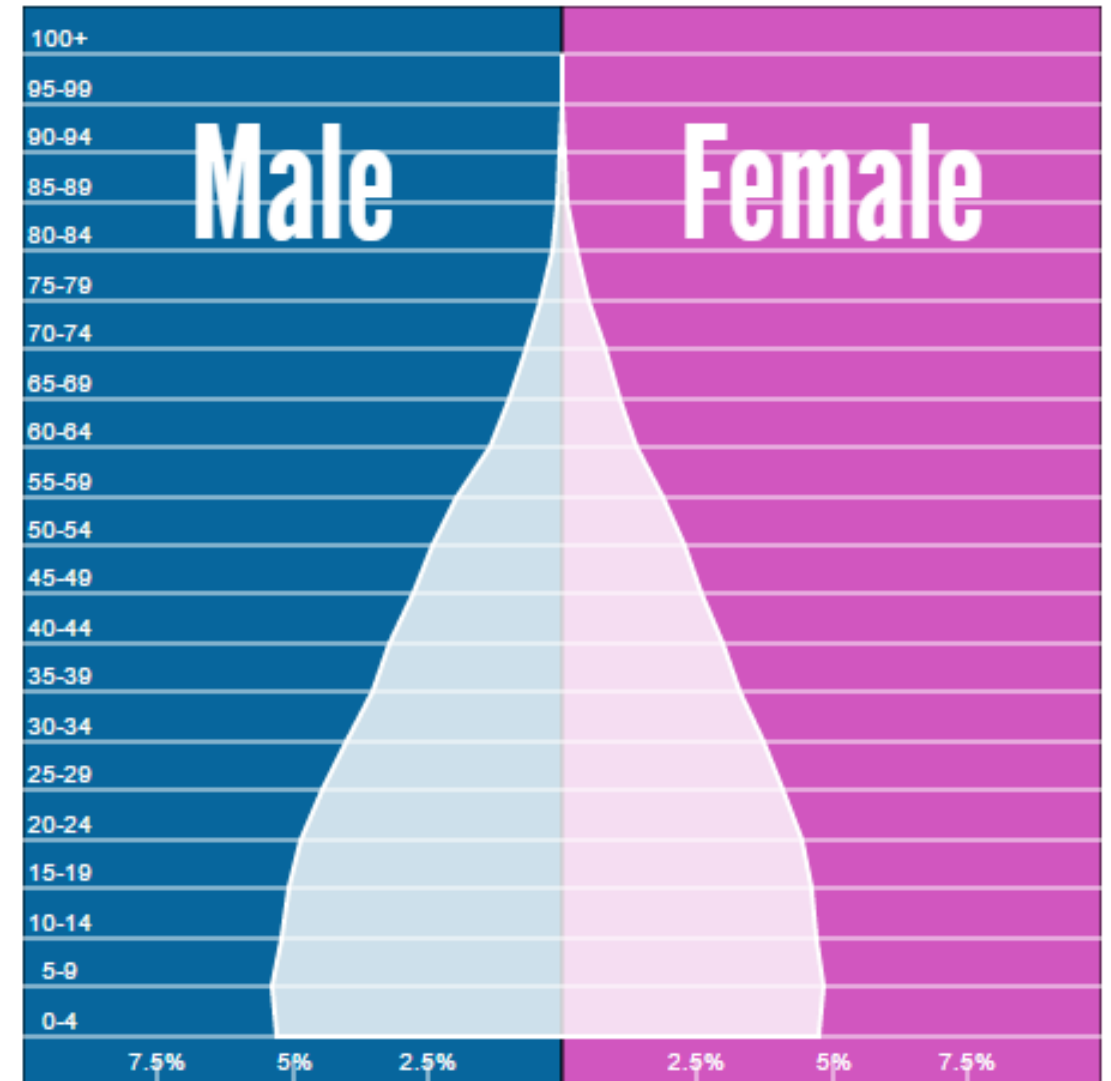
Gender

Religion

Caste

Linguistic group

INDIA- POPULATION PYRAMID



Link to this graph: <http://populationpyramid.net/india/>

CLASS ACTIVITY- READ AND DISCUSS

Demography in service of 'other' social sciences

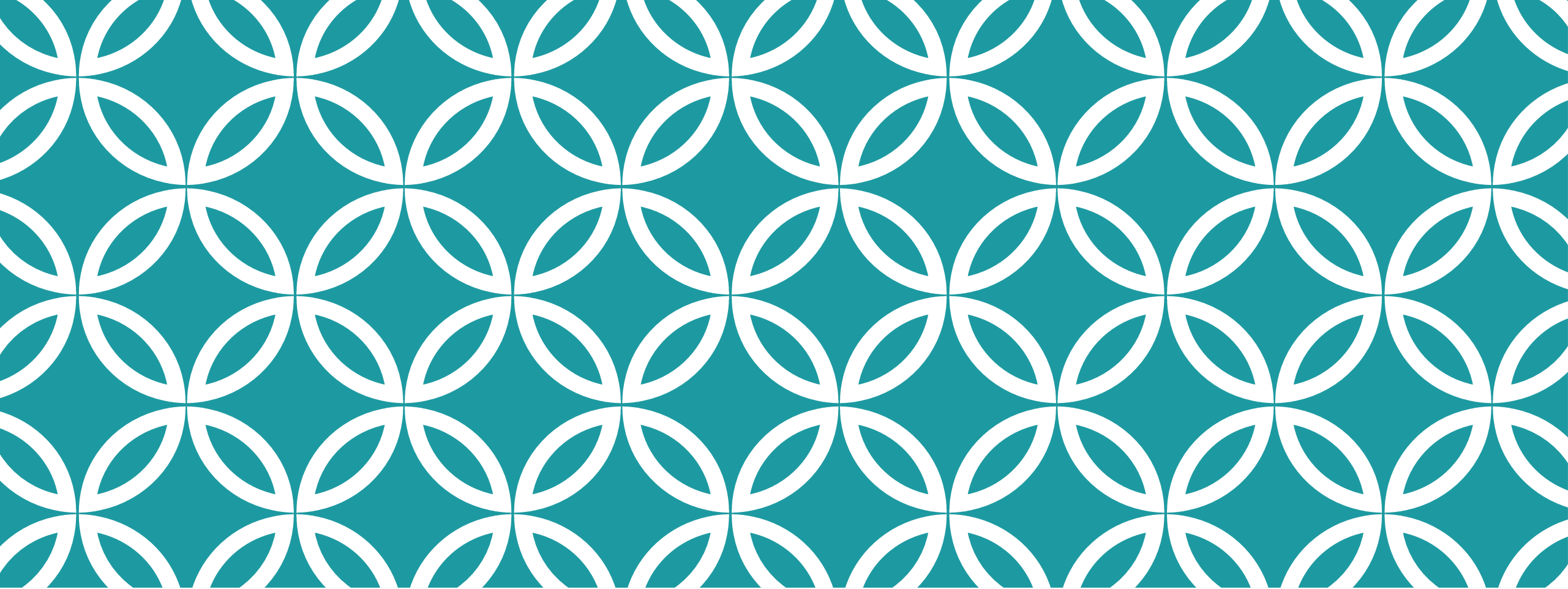
Shivakumar Jolad, May 22, 2023

<https://shiva-jolad-46811.medium.com/demography-in-service-of-other-social-sciences-6cfef9fae568>

Why do we need to educate people about the population?

Shivakumar Jolad 12 min read Jun 23, 2023

<https://shiva-jolad-46811.medium.com/why-do-we-need-to-educate-people-about-the-population-5ca41e1e43fc>



DEMOGRAPHY AT LARGE |

WORLD POPULATION

The UN estimated that the 8 billionth person (baby) was born in Nov 15, 2022. The life that this person depends on where they were born.

Born in a world facing global population growth, and yet a world in which some countries have unprecedented **population ageing** and face imminent decline.

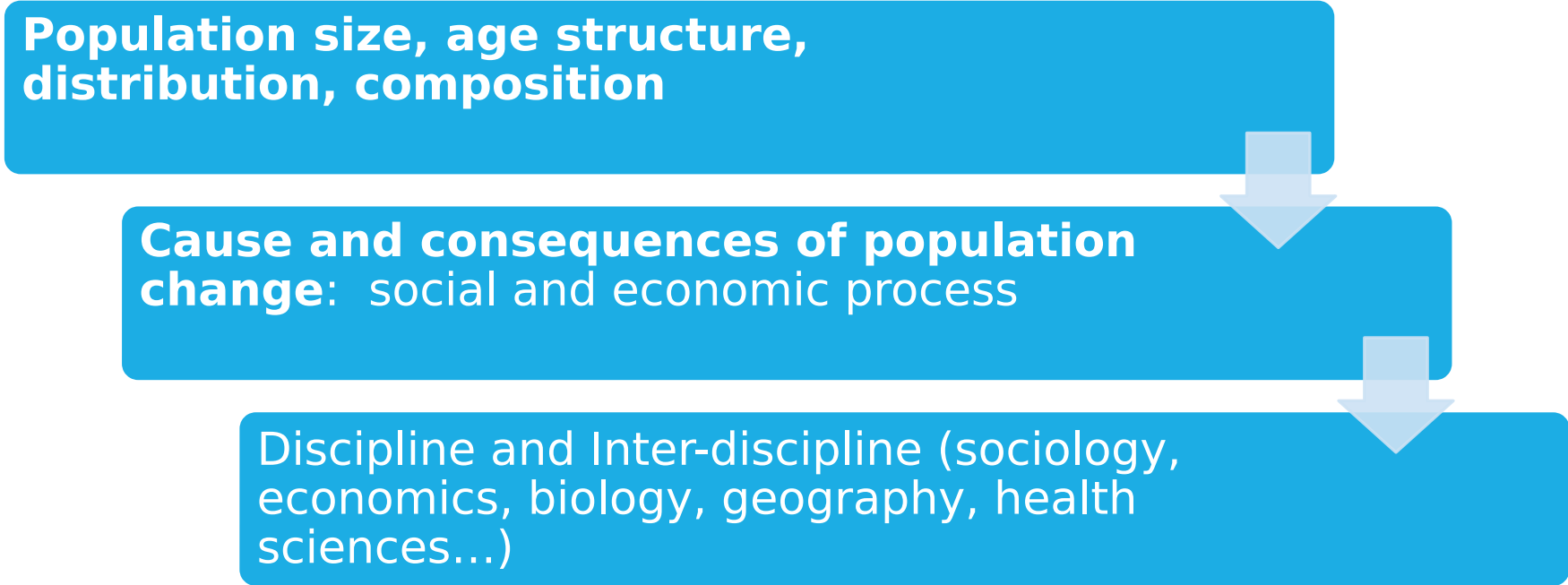
In a country with a life expectancy of 80 years. Or they may have a life expectancy of 45 years.

In a country with less than 1% of the population are HIV+, or a country where over 20% of the population are.

DEMOGRAPHY

Scientific study of population structure , change and its process

Population size, age structure, distribution, composition



Cause and consequences of population change: social and economic process

Discipline and Inter-discipline (sociology, economics, biology, geography, health sciences...)

McFalls, Joseph A. *Population: a lively introduction*. Vol. 58, no. 4. Population Reference Bureau, 2003.

QUESTIONS DEMOGRAPHY ADDRESSES

Is population growth a benefit or a burden? Can we feed the world?

Why do people in developing countries have more children than those in the developed societies?

Why do Japanese, on an average, live longer than Afghans?

How does sex-ratio affect marriage chances ?

Can Migration address the 'population problem' of Europe?

Can young population *growth* explain growth of Asian Tiger economies?

Will developed countries experience near zero population growth? What are the social implications?

Adapted from: Population Analysis for Policies & Programmes, UNFPA and IUSSP,
and from Tannistha Samanta discussions

SYLLABUS: BROAD CATEGORIES

Demographic Analysis: The study of components of variation and change in demographic variables and the relationships between them. (Formal/Mathematical Demography)

Population Process: The study of the relationships between demographic variables and other variables such as social and economic variables
(Social Demography)

COURSE TOPICS

<u>Sl.No</u>	<i>Topics</i>	<i>Hours</i>
1	Introduction to Demography: Origins of Demography, Components of population change.	3
2	Basic population measures, age-sex composition structure, population pyramids.	4
3	Mortality measures and Life table	5
4	Measures of Fertility and fertility change	5
5	Measures of Migration: National and Internal Migration	3
6	Population projection methods	4
7	Demographic Data: Cross sectional and longitudinal surveys	3
8	Demographic Theories and Perspectives	4
9	Socioeconomic cause of fertility change	3
10	Demographic methods in Public health	4
11	Migration and Urbanization	4
12	Forecasting future scenarios of Population and society	3
	TOTAL	45

Form Number

Census of India 2011

Household Schedule

Confidential when filled

Use only arabic numbers as indicated here

0123456789

SIDE-A

Start Here

Location Particulars

State/UT

District

Tahsil/Taluk/ P.S./Dev. Block Circle/Mandal

Town/ Village

Ward Code No. (only for Town)

Enumeration Block Number & Sub-Block No.

To be copied from Abridged Household

Household Block Number (Column 2 of section 2)

Household Number (Column 6 of section 2)

Serial Number of Household (Column 8 of section 2 or 3 or column 6 of section 4)

Type of Household:

Normal1

Institutional ..2

Houseless3

If institutional household, give details

Serial number	Q. 1 Name of the person start with head of household	Q. 2 Relationship to head write the relationship in full	Q. 3 Sex Male ...1 Female 2 In case the respondent wishes to return other than code 1 or 2 then give code '3' Other...3	Q. 4 Date of birth and Age 4(a) Date of birth as per English calendar (as declared or estimated) Day - Month - Year 4(b) Age Also write age on last birthday in completed years, in box against 4(b)	Q. 5 Current marital status give code from list below	Q. 6 Age at marriage In completed years (not applicable for Never married)	Q. 7 Religion (Write name of the religion in full) Also give code in box if found in the list below For other religions, write name of the religion in full but do not give any code number	Q. 8 * Scheduled Caste (SC)/ Scheduled Tribe (ST) 8(a) Is this person SC/ST? If 'YES' give code in box SC1 ST2 If 'NO' put '3' in box 8(b) If SC or ST write name of the SC or ST from the list supplied.	Q. 9 Disability 9(a) Is this person mentally / physically disabled? Yes-1/No-2 9(b) If 'Yes' in 9(a), give code in the box against 9(b) from the list below 9(c) If 'multiple disability' (Code '8' in 9(b), give maximum three codes in boxes against 9(c) from the list below	Q. 10 Mother tongue write name of the mother tongue in full.	Q. 11 Other languages known write upto two languages in order of proficiency excluding mother tongue	Q. 12 Literacy status Literate ...1 Illiterate ..2 M F O	Q. 13 Status of attendance in educational institution give code from list below	Q. 14 Highest educational level attained Write the full description. For diploma or degree holder, also write the subject of specialisation.
				4(a) D D - M M Y Y Y Y 4(b)				8(a) 8(b)	9(a) 9(b) 9(c)					
				4(a) D D - M M Y Y Y Y 4(b)					9(a) 9(b) 9(c)					
				4(a) D D - M M Y Y Y Y 4(b)					9(a) 9(b) 9(c)					

CENSUS HOUSEHOLD SCHEDULE PART A

17

Census of India 2011		Household Schedule										Confidential when filled		Use only arabic numbers as indicated here										SIDE- B	
Q. 1	Q. 15	Q. 16	Q. 17	Q. 18			Q. 19	Q. 20	Q. 21	Q. 22	Q. 23	Q. 24			Q. 25	Q. 26	Q. 27	Q. 28	Q. 29						
Serial number	Name of the person	Worked any time during last year Include part time help, unpaid work on farm, family enterprise or in any other economic activity	Category of economic activity Fill for main or marginal worker (If code '1' or '2' or '3' in Q.15)	Characteristics of workers and non-workers				Migration characteristics				Fertility particulars													
				Fill for worker in household industry or other worker (If code '3' or '4' in Q.16)	Fill for marginal worker or non worker (If code '2' or '3' or '4' in Q.15)	Fill for other worker (If code '4' in Q.16)	Fill for person born outside this village/town	Fill for person who has come to this village/town from elsewhere	Fill for currently married, widowed, divorced or separated woman	Fill for currently married woman															
Copy from side A in same order				Occupation Describe the actual work	Nature of industry, trade or service where the person works/ worked or self employed.	Class of worker give code from list below	Non-economic activity give code from list below	Seeking or available for work Yes ..1 No ...2	22(a) One-way distance from residence to place of work in kilometres 22(b) Mode of travel to place of work give code from list below	22(a) One-way distance from residence to place of work in kilometres 22(b) Mode of travel to place of work give code from list below	Birth place if within India, write the present name of the village/town, district and state. Also write '1' for village or '2' for town in the box. if outside India, write the present name of the country and put '-' against village/town and district.	24(a) if within India, write the present name of the village/town, district and state. Also write '1' for village or '2' for town in the box. if outside India, write the present name of the country and put '-' against village/town and district.	24(b) at the time of migration Rural...1 Urban...2	Reason for migration Work/Employment 1 Business2 Education3 Marriage4 Moved after birth 5 Moved with household6 Any other7	Duration of stay in this village/town since migration in completed years (write '00' if less than a year)	Children surviving No. of children surviving at present	Children ever born Total no. of children ever born alive	Number of Children born alive during last one year (1st March 2010 to 28th February 2011)							
									a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country						Daughter Son	Daughter Son	Daughter Son						
									a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country						Daughter Son	Daughter Son	Daughter Son						
									a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country						Daughter Son	Daughter Son	Daughter Son						
									a) distance in km. b) mode of travel	Village/Town District State/Country	Village/Town District State/Country						Daughter Son	Daughter Son	Daughter Son						

HOME- ASSIGNMENT CENSUS HOUSEHOLD SCHEDULE PART B

ACTIVITY- CLASS DEMOGRAPHY

Name	Age	Gender	Place of birth (City/District, State/Country)	Educatio n level	Mother Tongue	Religion

TEXT BOOKS/REF ERENCES

Population Analysis for Policies & Programmes.
by IUSSP and UNFPA -
<http://papp.iussp.org/index.html>

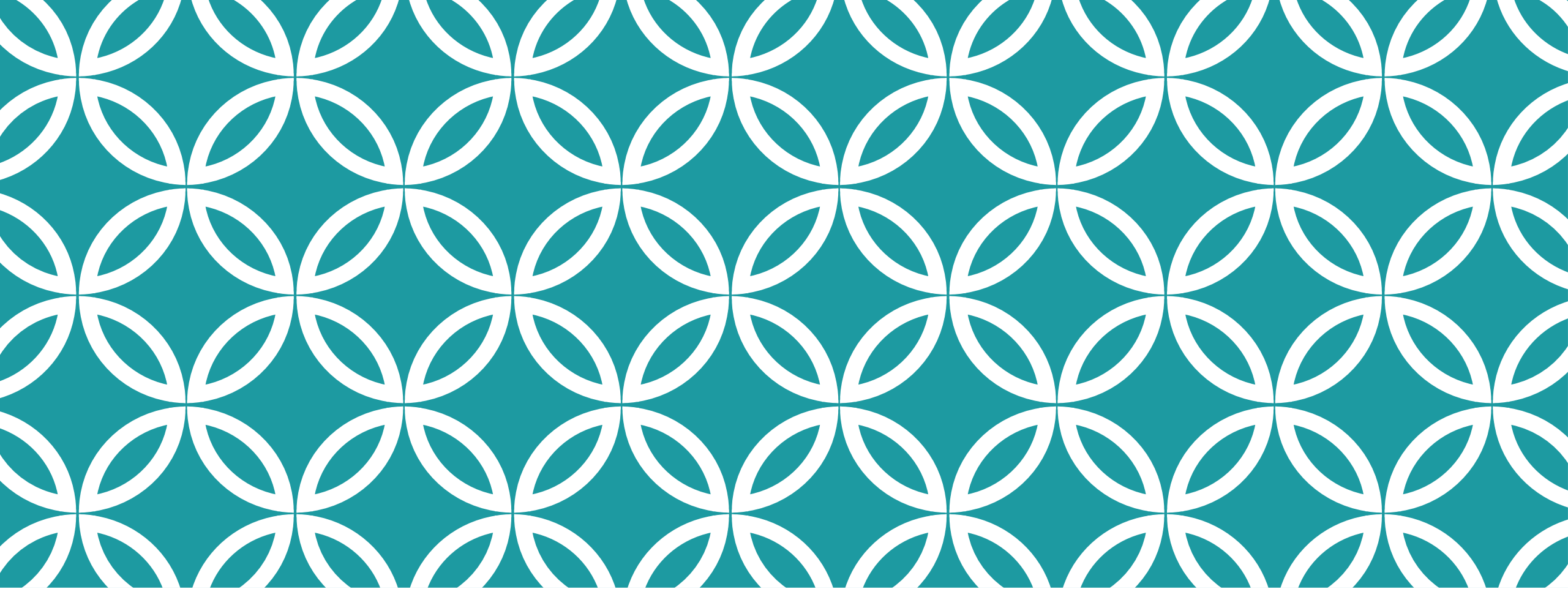
(Poston and Bouvier, 2010) **POPULATION AND SOCIETY: An Introduction to Demography** Dudley L. Poston, Jr., Leon F.

Preston Samuel H, Heuveline Patrick, Guillot Michel.
Demography: Measuring and Modeling Population Processes. Maiden, MA: Blackwell; 2001.

Rowland, Donald T. *Demographic Methods and Concepts*. OUP Catalogue, Oxford University Press, 2003.

Schoen, Robert. *Modeling multigroup populations*. Springer, 1987.

Books on Social Demography, Review articles from Population reference bureau, UN Human Development Report, UNFPA and RAND corporation.



POPULATION GROWTH- WORLD |

HOW MANY PEOPLE EVER LIVED ON EARTH?

How many people
have ever lived?

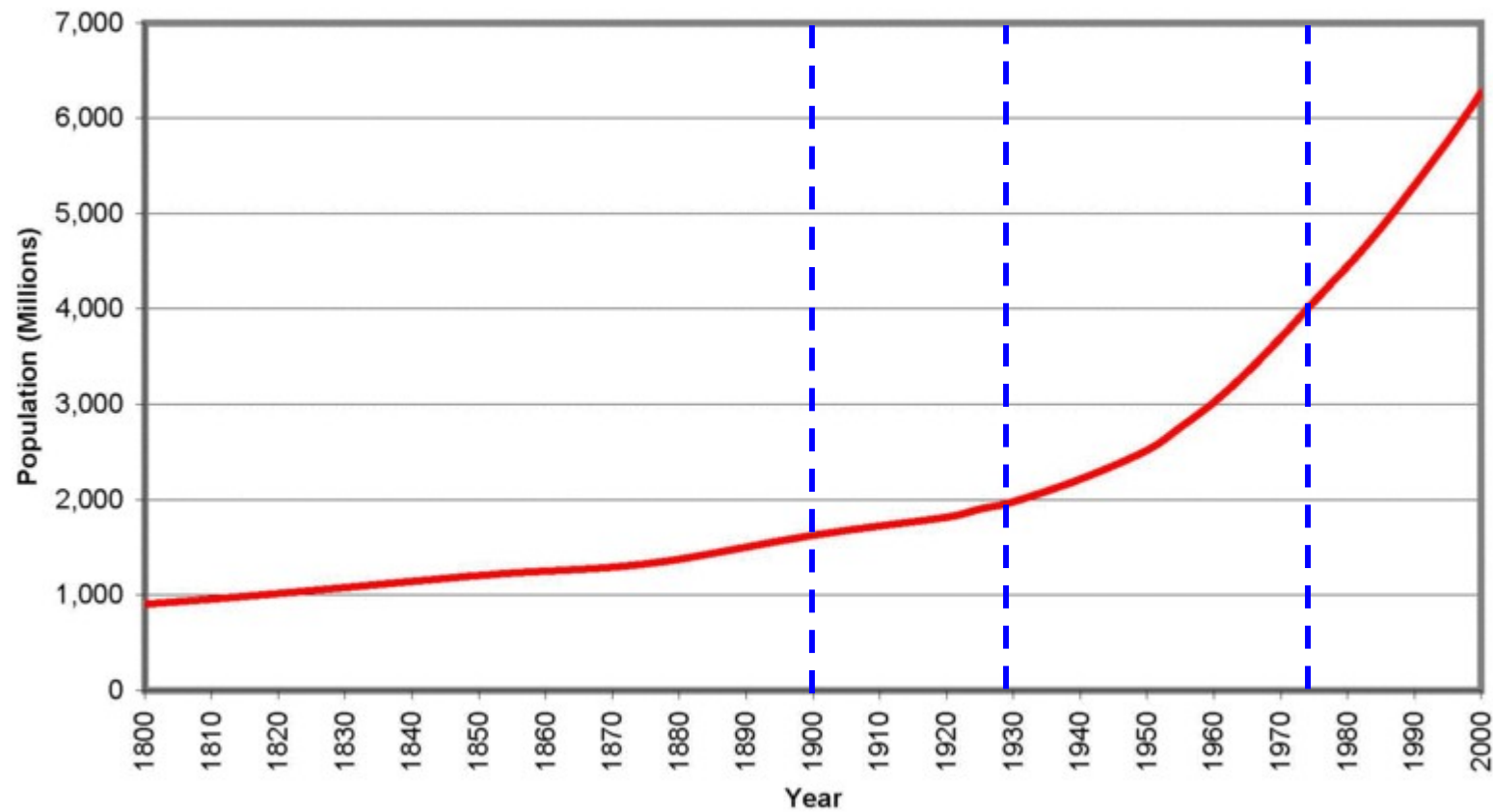
YEAR	POPULATION	BIRTHS PER 1,000	BIRTHS BETWEEN BENCHMARKS	NUMBER EVER BORN	PERCENT OF THOSE EVER BORN
190,000 B.C.E	2	80	–	–	–
50,000 B.C.E.	2,000,000	80	7,856,100,000	7,856,100,002	0
8000 B.C.E.	5,000,000	80	1,137,789,769	8,993,889,771	0.1
1 C.E.	300,000,000	80	46,025,332,354	55,019,222,125	0.5
1200	450,000,000	60	26,591,343,000	81,610,565,125	0.6
1650	500,000,000	60	12,782,002,453	94,392,567,578	0.5
1750	795,000,000	50	3,171,931,513	97,564,499,091	0.8
1850	1,265,000,000	40	4,046,240,009	101,610,739,100	1.2
1900	1,656,000,000	40	2,900,237,856	104,510,976,956	1.6
1950	2,499,000,000	31-38	3,390,198,215	107,901,175,171	2.3
2000	6,149,000,000	22	6,064,994,884	113,966,170,055	5.4
2010	6,986,000,000	20	1,364,003,405	115,330,173,460	6.1
2022	7,963,500,000	17	1,690,275,115	117,020,448,575	6.8
2035	8,899,000,000	16	1,758,578,889	118,779,027,464	7.5
2050	9,752,000,000	14	2,068,409,608	120,847,437,072	8.1

Sources as of November 2022: Toshiko Kaneda, Charlotte Greenbaum, and Carl Haub, *2022 World Population Data Sheet* (Washington, DC: Population Reference Bureau, 2022); United Nations, Department of Economic and Social Affairs, *World Population Prospects: The 2022 Revision* (New York: United Nations, 2022); personal communication with Dudley L. Poston Jr., Professor of Sociology and the George T. and Gladys H. Abell

Professor of Liberal Arts, Texas A&M University.

<https://www.prb.org/articles/how-many-people-have-ever-lived-on-earth/>

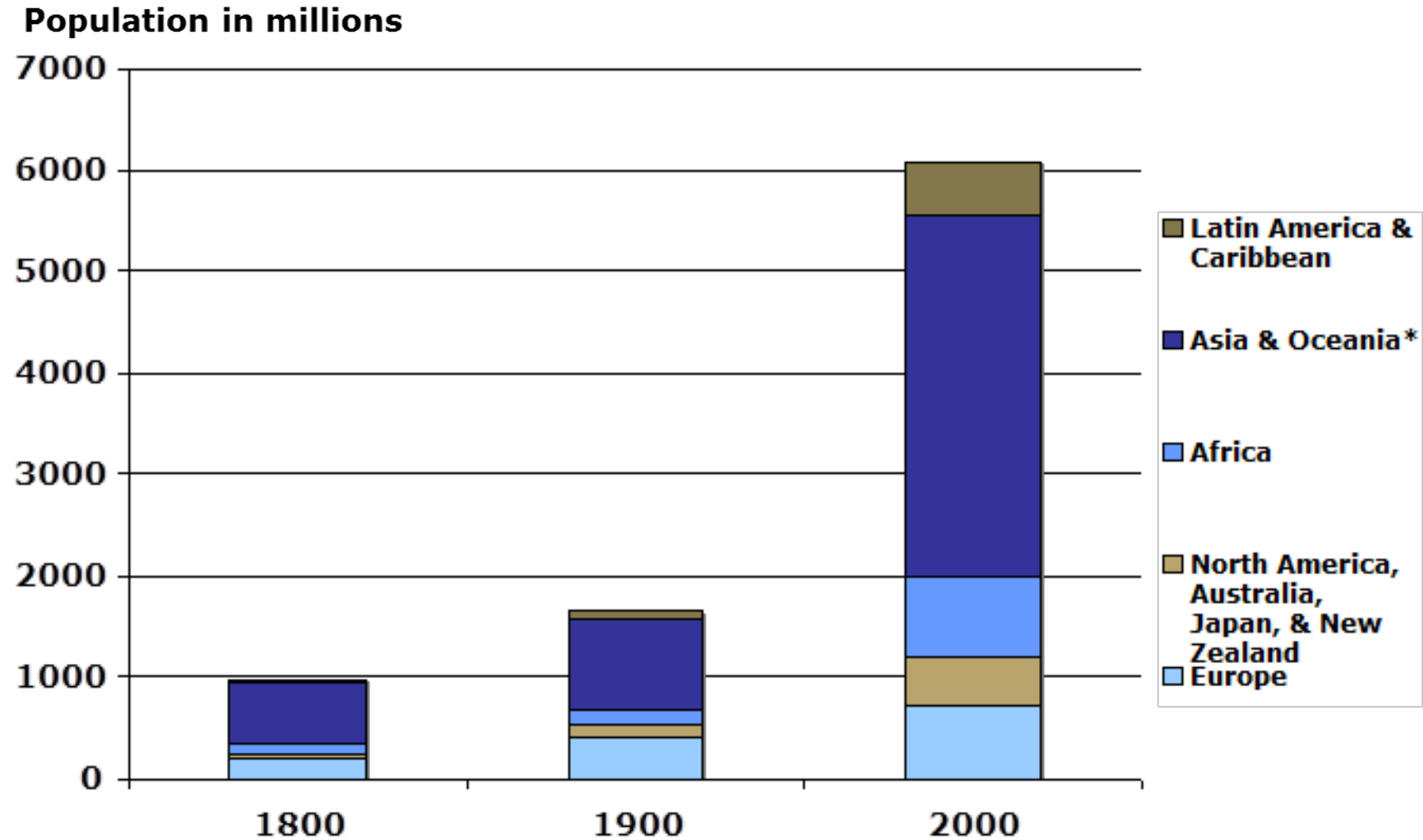
World Population, 1800 - 2000



Source: J. Bradford DeLong, "Estimating World GDP, One Million B.C. - Present." http://www.j-bradford-delong.net/TCEH/1998_Draft/World_GDP/Estimating_World_GDP.html. Accessed Mar 5, 2008.

© 2008. Michael W. Kruse

WORLD POPULATION- THREE CENTURIES

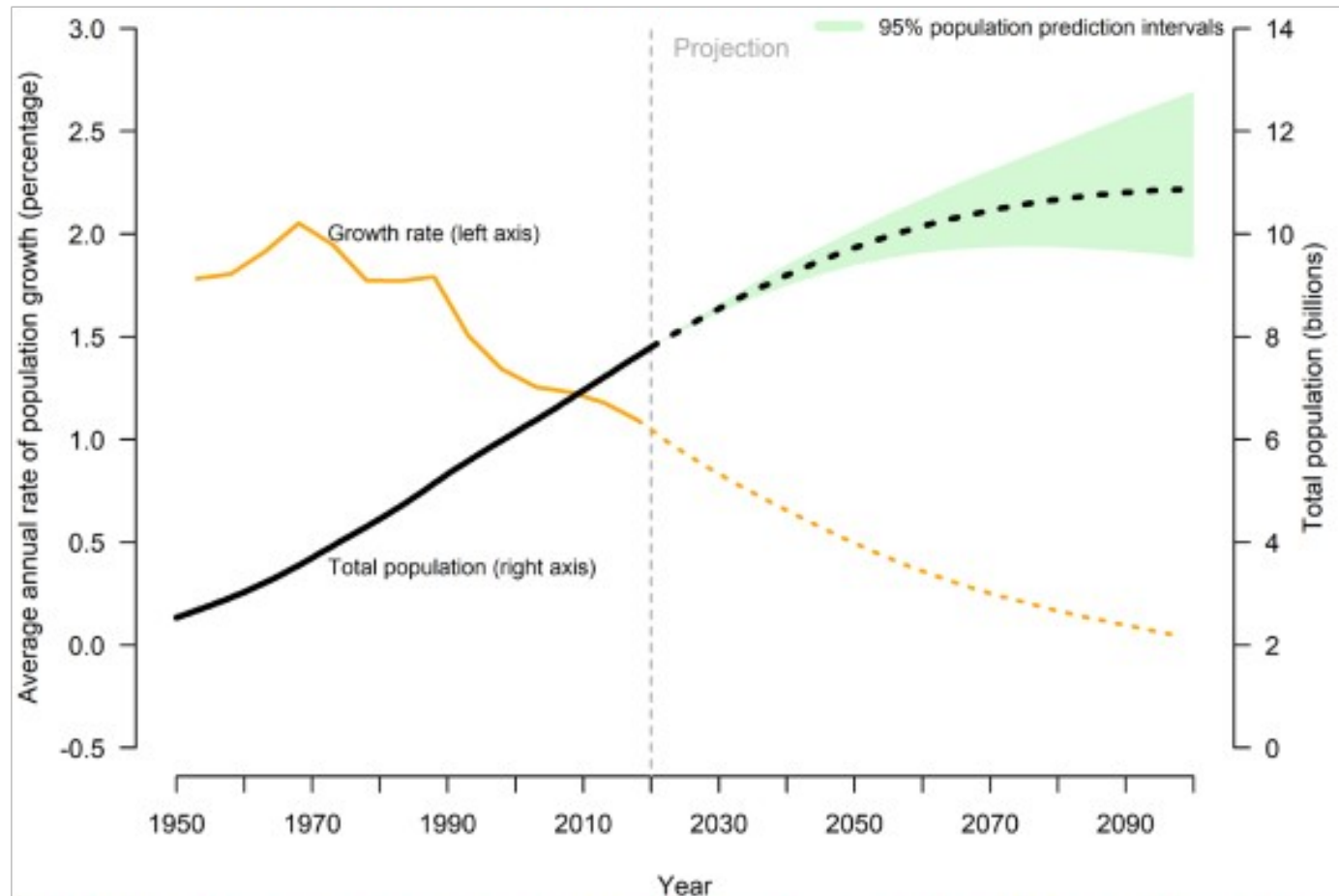


Sources: UN Population Division, *World Population Prospects: The 2004 Revision* (2005) and *Briefing Packet, 1998 World Population Prospects* (October 1998); and I.B. Taeuber, *The Population of Japan* (1958): 21-23.

WORLD POPULATION AND GROWTH RATE

Figure 1. Population size and annual growth rate for the world: estimates, 1950-2020, and medium-variant projection with 95 per cent prediction intervals, 2020-2100

Population growth continues at the global level, but the rate of increase is slowing, and the world's population could cease to grow around the end of the century



Data source: United Nations, Department of Economic and Social Affairs, Population Division (2019). *World Population Prospects 2019*.

Table 1. Population of the world, SDG regions and selected groups of countries, 2019, 2030, 2050 and 2100, according to the medium-variant projection

Region	Population (millions)			
	2019	2030	2050	2100
World	7 713	8 548	9 735	10 875
Sub-Saharan Africa	1 066	1 400	2 118	3 775
Northern Africa and Western Asia	517	609	754	924
Central and Southern Asia	1 991	2 227	2 496	2 334
Eastern and South-Eastern Asia	2 335	2 427	2 411	1 967
Latin America and the Caribbean	648	706	762	680
Australia/New Zealand	30	33	38	49
Oceania*	12	15	19	26
Europe and Northern America	1 114	1 132	1 136	1 120
Least developed countries	1 033	1 314	1 877	3 047
Land-locked Developing Countries	521	659	926	1 406
Small Island Developing States	71	78	87	88

Data source: United Nations, Department of Economic and Social Affairs, Population Division (2019). *World Population Prospects 2019*.

* excluding Australia and New Zealand

World Population Prospects 2019

Highlights

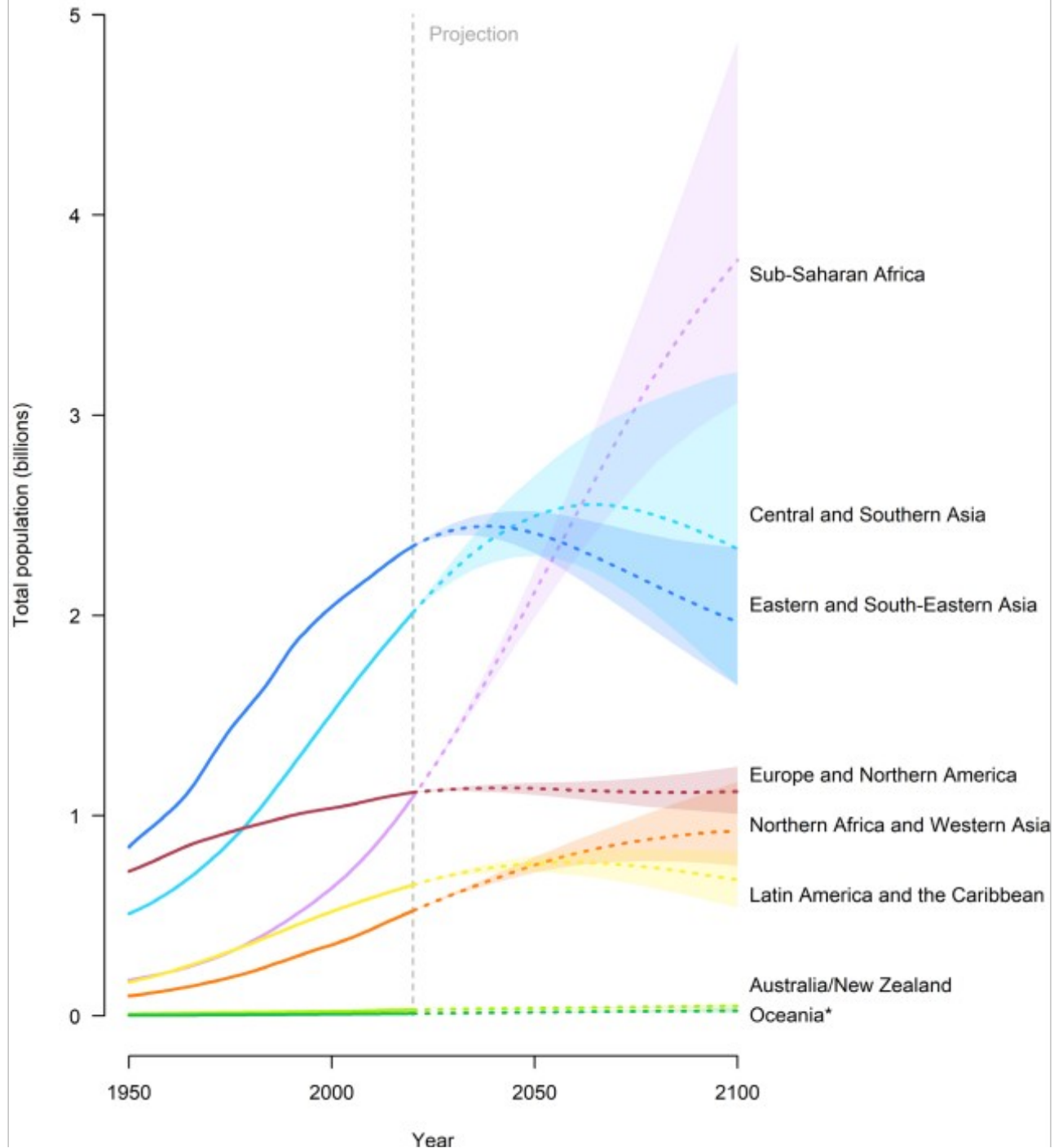


Figure 2. Population by SDG region: estimates, 1950-2020, and medium-variant projection with 95 per cent prediction intervals, 2020-2100

Of the eight SDG regions, only sub-Saharan Africa is projected to sustain rapid population growth through the end of the century, according to the medium-variant projection

World Population Prospects 2019

Highlights

DEMOGRAPHIC ANALYSIS

**Thinking about populations:
age and sex**

Measure demographic events:

- Measurement of mortality and age standardization
- Measuring fertility. Proximate determinants

Life tables , Model Life tables

**Migration and population
distribution**

**Population projections:
concepts and methods**

**Population dynamics and
momentum**

MINI QUIZ

Child mortality is a measure of health of population (broadly)

- Correlates strongly with functioning and utilization of public health services

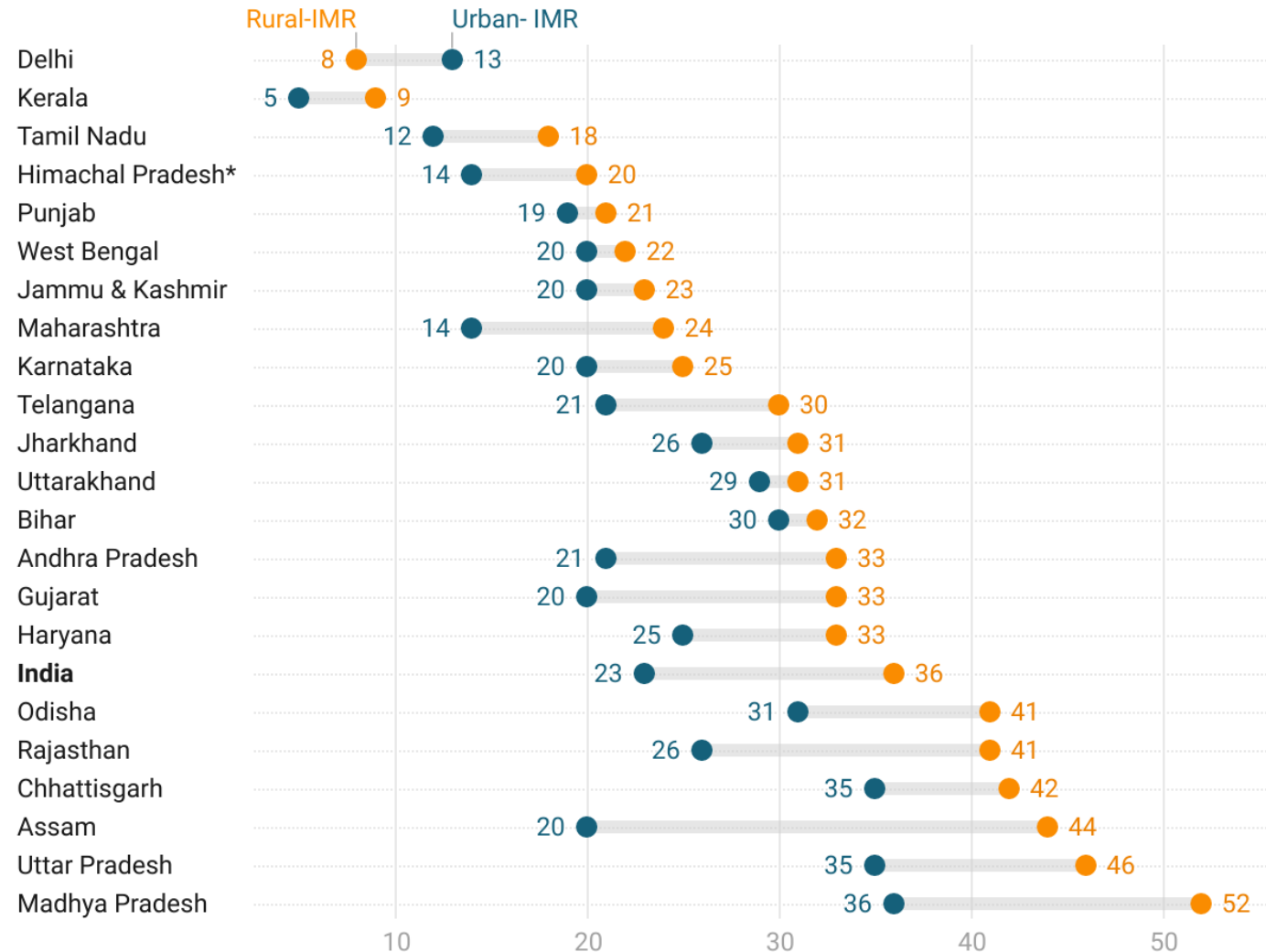
Among the **paired states** which one has **lower child (Infant) mortality?**

	State 1	State 2
1	Kerala	Karnataka
2	Andhra Pradesh	Mizoram
3	Gujarat	Himachal Pradesh
4	Uttar Pradesh	West Bengal
5	Uttaranchal	Assam

INFANT MORTALITY BY STATES

SRS-
2018

Infant Mortality rate Indian States (per 1000 live births)



Source: Sample Registration System- 2018 • Created with Datawrapper

REDUCING CHILD MORTALITY LEADS TO OVERPOPULATION?



Hans Rosling, Gapminder
<https://www.youtube.com/watch?v=BkSO9pOVpRM>